QQ-W-345b
June 10, 1965
SUPERSEDING
Int. Fed Spec. QQ-W-00345a(GSA-FSS)
March 16, 1964 and
Fed. Spec. QQ-W-345
July 23, 1958

FEDERAL SPECIFICATION

WIRE, ELECTRICAL, STEEL, COPPER-COVERED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers copper-covered steel wire for telephone, telegraph, power transmission, and antenna applications where high tensile strength is advantageous.

1.2 Classification.

1.2.1 *Types.* Copper-covered steel wire furnished under this specification shall be an integral composite product of the following types, as specified (see 6.1).

Type I—High strength, 40-percent conductivity (ASTM grade 40 HS).

Type II—High strength, 30-percent conductivity (ASTM grade 30 HS).

Type III—Extra-high-strength, 30-percent conductivity (ASTM grade 30 EHS).

Type IV—Extra-high-strength, 40-percent conductivity (ASTM grade 40 EHS).

1.2.2 *Sizes.* The wire furnished under this specification shall be of the sizes shown in table I, as specified (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

UU-P-271—Paper, Wrapping, Waterproofed Kraft.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

TABLE I. *Tensile requirements*

Nominal diameter	Tensile strength, minimum				Elongation minimum
	Type I	Type II	Type III	Type IV	All types
Inch	*P. s. i.*	*P. s. i.*	*P. s. i.*	*P. s. i.*	*Percent*
0.2043	108,000	120,000	142,500	.....	1.5
.1819	113,000	125,000	150,500	.....	1.5
.1650	118,000	130,000	157,500	.....	1.5
.1620	118,000	130,000	157,500	.....	1.5
.1443	123,000	135,000	164,000	.....	1.5
.1285	128,000	140,000	170,000	.....	1.5
.1280	128,000	140,000	170,000	.....	1.5
.1144	133,000	145,000	174,100	.....	1.5
.1040	138,600	151,000	175,000	156,000	1.5
.1019	138,600	151,000	179,000	.....	1.5
.0808	153,100	.....	.....	.....	1.5
.0800	153,200	.....	179,000	.....	1.5
.0720	120,000	130,000	.....	.....	1.0
.0641	124,000	136,000	170,000	.....	1.0
.0571	117,000	129,000	.....	.....	1.0
.0508	123,000	133,000	.....	.....	1.0
.0453	115,000	149,000	.....	.....	1.0
.0403	120,000	157,000	.....	.....	1.0
.0359	121,000	133,000	.....	.....	1.0
.0320	125,000	137,000	.....	.....	1.0
.0285	115,000	127,000	.....	.....	1.0
.0253	115,000	127,000	.....	.....	1.0
.0226	115,000	127,000	.....	.....	1.0
.0201	115,000	127,000	.....	.....	1.0

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Military Specification:

MIL-L-10547—Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Standards Association, Inc. (ASA) Publication:

C7.17—Hard-drawn Copper Covered Steel Wire, Specifications for (ASTM B227)

(Copies may be obtained from American Standards Association, Inc., 10 East 40th Street, New York, N.Y., 10016).

3. REQUIREMENTS

3.1 General. Except as otherwise provided herein, copper-covered steel wire covered by this specification shall conform to the requirements of ASA C7.17.

3.2 Mechanical requirements.

3.2.1 *Tensile strength and elongation.* The tensile strength and elongation of copper-covered steel wire shall conform to the requirements in table I. For sizes 0.080 to 0.2043 inch in diameter, wire having a nominal diameter which is more than 0.001 inch greater than a size in this range listed in table I shall conform to the tensile requirements of the next larger size. For sizes 0.020 to 0.079 inch in diameter, wire having a diameter other than the sizes listed in table I shall conform to the tensile requirements of the nearest listed size.

3.2.2 *Torsion requirements.* The finished wire shall withstand not less than 20 twists without fracture for wire 0.080-inch diameter and larger and not less than 25 twists for wire 0.020-inch to 0.079-inch diameter when subjected to the torsion test of ASA C7.17. When twisted to destruction, the wire shall not reveal any seams, pits, slivers, or surface imperfections.

3.3 Thickness of copper covering. The thickness of copper at any point on the wire shall be not less than shown in table II.

TABLE II. *Thickness of copper covering*

Nominal diameter	Thickness of copper-minimum	
	Types I and IV	Types II and III
Inch	*Inch*	*Inch*
0.200 to .209	0.012	0.010
.180 to .199	.011	.009
.160 to .179	.010	.008
.150 to .159	.009	.007
.140 to .149	.008	.007
.130 to .139	.008	.006
.120 to .129	.007	.006
.110 to .119	.006	.005

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TABLE II. *Thickness of copper covering —(Continued)*

Nominal diameter	Thickness of copper-minimum	
	Types I and IV	Types II and III
Inch	*Inch*	*Inch*
.100 to .109	.006	.005
.080 to .099	.005	.004
.070 to .079	.004	.003
.060 to .069	.003	.003
.050 to .059	.003	.002
.040 to .049	.002	.002
.030 to .039	.001	.001
.020 to .029	.001	.001

3.4 Dimensional requirements. The wire shall be circular in cross-section and permissible variation from the nominal diameters listed in table I shall not exceed the following limits:

Nominal wire diameter	Permissible variation
0.100 inch and larger	Plus or minus 1-1/2 percent
0.060 to 0.099 inch	Plus or minus 0.0015 inch
0.020 to 0.059 inch	Plus or minus 0.001 inch

3.5 Machine-straightened wire. When so specified in the invitation for bids (see 6.1), the wire shall be machine-straightened.

3.6 Workmanship. The wire shall be manufactured and finished in a thoroughly workmanlike manner, free from injurious defects, and shall contain no visible splits, nicks, or slivers.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.1), all measurements shall be made at room ambient temperature, pressure, and humidity.

4.3 Sampling.

4.3.1 For visual and dimensional examinations, the sample size shall be selected in accordance with MIL-STD-105, inspection level II, with acceptance based on an acceptable quality level (AQL) of 6.5 defects per hundred units.

4.3.2 For tests, the sample size shall be selected in accordance with MIL-STD-105, inspection level S-2, with acceptance based on an AQL of 6.5 defects per hundred units.

4.4 Inspection and tests. Except as otherwise provided herein, inspection and tests shall be in accordance with ASA C7.17.

4.4.1 *Elongation determination.* Unless otherwise specified (see 6.1), elongation of the wire shall be determined by measure-

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ments made between the jaws of the tensile testing machine.

4.4.2 *Machine-straightening test.* When machine-straightened wire is specified, the effectiveness of the machine-straightening operation shall be determined by unwinding a length of four complete loops from a coil of such wire and pulling it out straight. When the free end is released, the wire shall lie flat on the floor and the diameter of the ring which is formed at the end shall be not less than 5½ feet.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine compliance with packaging, packing, and marking requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of levels of preservation, packaging, and packing shall be in accordance with Fed. Std. No. 102).

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A or B.*

5.1.1.1 *Unit packaging.*

5.1.1.1.1 *Coils (AWG sizes 8 to AWG 24).* Cable and wire shall be uniformly coiled and packaged 500 feet in a fiberboard box conforming to PPP-B-636, type CF or SF provided with the special knock-out feature normally used in commercial practice for dispensing purposes.

5.1.1.1.2 *Coils (AWG sizes 4 to AWG 7).* Cable and wire shall be uniformly coiled and packaged 500 feet per coil. The coils shall be spirally wrapped with waterproof kraft paper conforming to UU-P-271, class E-2 and sealed.

5.1.1.1.3 *Reels.* Cable and wire shall be packaged on nonreturnable wood or metal spools or reels suitably designed with strong flanges of sufficient diameter to protect the cable or wire from damage during shipment and storage. After the cable or wire has been wound on the reel, it shall be covered with a full wrap of waterproof kraft barrier material conforming to UU-P-271, class E-2. The wrap shall extend between the flanges and be secured in place with a minimum of two flat steel straps.

5.1.2 *Level C.* Cable and wire shall be packaged in coils or on reels of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Coils (AWG sizes 8 to AWG 24).* Cable and wire packaged as specified in 5.1.1.1.1 shall be packed in nailed wood or wood cleated plywood boxes conforming to PPP-B-621 and PPP-B-601, respectively. Containers shall be fitted with waterproof case liners conforming to MIL-L-10547, class 2, and sealed in accordance with appendix thereto.

5.2.1.2 *Coils (AWG sizes 4 to AWG 7) and reels.* Cable and wire packaged as specified in 5.1.1.1.2 and 5.1.1.1.3 shall require no overpacking.

5.2.2 *Level B.*

5.2.2.1 *Coils and reels.* Cable and wire packaged as specified in 5.1.1.1 shall require no overpacking.

5.2.3 *Level C.* The subject commodity shall be packaged and packed in substantial containers of the type, size, and kind commonly used for the purpose, so constructed

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as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Type of wire (1.2.1).

(*c*) Size of wire (1.2.2).

(*d*) Quantity of wire.

(*e*) Machine-straightened wire, if required (3.5).

(*f*) Inspection responsibility, if other than specified (4.1).

(*g*) Inspection conditions, if other than specified (4.2).

(*h*) Method of measuring elongation, if other than specified (4.4.1).

(*i*) Level of packaging and packing required (5.1 and 5.2).

(*j*) Special marking, if required (5.3).

6.2.1 *Percent conductivity.*

6.2 **Definitions.**

6.2.1.1 *Forty percent conductivity.* At power frequencies or direct current, the conductance of types I and IV wire is approximately 40 percent of that of a hard-drawn copper wire of the same size.

6.2.1.2 *Thirty percent conductivity.* At power frequencies or direct current, the conductance of types II and III wire is approximately 30 percent of that of a hard-drawn copper wire of the same size.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Wire, copper clad steel.

Carload minimum weight 30,000 pounds.

Motor:

Wire, copper clad steel.

Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:

Custodians:

Army—EL

Navy—YD

Air Force—17

Review:

Army—MU, MI

Navy—YD, SH

Air Force—17

NSA

DSA—IS

Coordinating Activity:

Army—EL

Users:

Army—MO

Navy—MC

Air Force—85

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